

Project Report: Household Services Application

1. Student Details

Name: Tushar Jalan

Roll Number: 23f2003751

Email: 23f2003751@ds.study.iitm.ac.in

2. Project Overview

Project Title: Household Services Application

Problem Statement:

To develop a platform that bridges the gap between customers seeking household services

(e.g., plumbing, cleaning, electrical repairs) and professionals offering these services. The platform should provide a smooth user experience for managing bookings, communication, and service fulfillment.

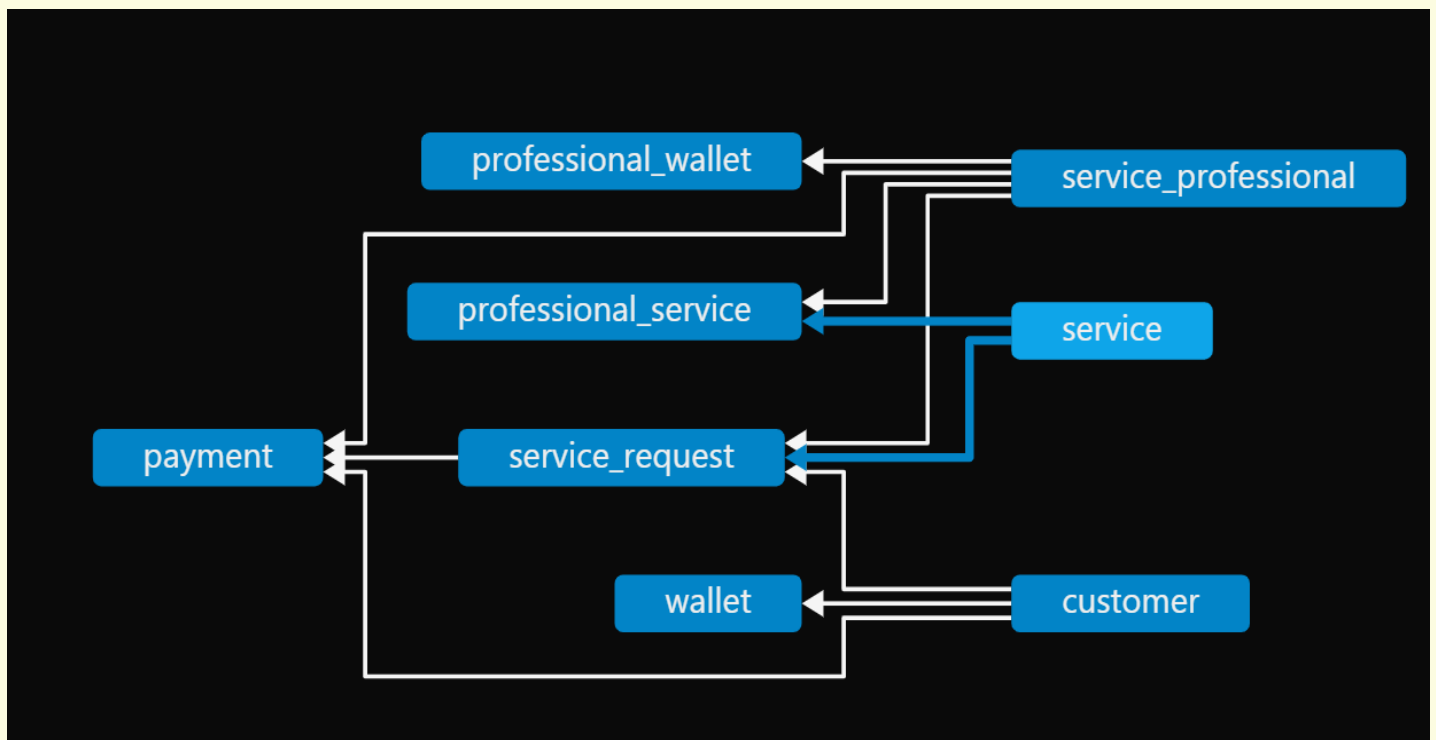
3. Approach

- Designed a multi-user architecture with distinct roles (Admin, Professionals, Customers).
- Developed user authentication and role-based authorization.
- Created customer-friendly interfaces for booking services and service tracking.
- Integrated admin functionality to manage service professionals and monitor activities.
- Implemented dashboards for professionals to manage requests and schedules.
- Added intuitive search system (by category, rating, pincode, etc.).

4. Frameworks and Libraries Used

- **Flask:** Backend framework for building web applications.
- **SQLAlchemy:** ORM (Object-Relational Mapping) tool for database interactions.
- **SQLite:** Database management system for storing application data.
- **HTML/CSS/JavaScript:** Frontend technologies for user interface design and interactivity.
- **Datetime:** Python library for handling date and time operations.
- **Jinja2:** Template engine for rendering dynamic HTML content.
- **ChartJS:** User for creating different types of charts on the admin dashboard

5. ER Diagram



6. API Resource Endpoints

- **Admin Summary API**
 - o **URL:** /admin/summary/api
 - o **Method:** GET
 - o **Description:** Fetches the customer ratings and service request summary for the admin.
- **Professional Dashboard Summary API**

- o **URL:** /professional_dashboard/summary/api
 - o **Method:** GET
 - o **Description:** Fetches the rating data and service status summary (including requested, accepted, rejected, and completed services) for the logged-in professional.
- **Service Request API**
 - o **URL:** /api/request_service
 - o **Method:** POST
 - o **Description:** Allows customers to request a service by selecting a professional who offers the service. If a professional is available, a service request is created and stored.
- **Professional Details API**
 - o **URL:** /api/professional/<int:professional_id>
 - o **Method:** GET
 - o **Description:** Fetches the details of a specific professional, including their name, experience, service type, and verified status.
- **Customer Dashboard Summary API**
 - o **URL:** /customer_dashboard/summary/api
 - o **Method:** GET
 - o **Description:** Fetches the ratings data and service status summary (including requested, accepted, rejected, and completed services) for the logged-in customer.

7. Presentation Video

Drive Link: https://drive.google.com/file/d/1dgWD5u_4DzkHOVewhee_gYE-uag-5f0S/view?usp=sharing

8. Conclusion

The project successfully developed a platform that connects household service providers with customers.

It is robust, user-friendly, and scalable, with the potential for future enhancements like online payment integration and real-time service tracking.